COORDINATE GEOMETRY AND VECTOR ANALYSIS

A visual study summary for MATH 201 / MATH 2201

Core concepts • essential formulas • worked examples • region and surface diagrams

What is improved in this edition?

- Added diagrams for vectors and projections, double-integral regions, polar coordinates, triple-integral solids, tangent planes, and surface normals.
- Kept the eight-topic sequence of the supplied study notes.
- Added setup strategies, common mistakes, worked examples, and a compact formula review.

Original explanations informed by standard multivariable-calculus methods in Anton, Bivens & Davis, Calculus: Early Transcendentals, and James Stewart's calculus texts. This is an independent study aid, not a reproduction of those books.

How to use this guide

For each problem, identify the geometric object first, sketch it, choose coordinates, set correct bounds, and only then calculate. A correct setup is often the hardest and most important part of multivariable calculus.

Study pass	Goal
1. Meaning	Explain what the quantity measures: length, area, volume, flux, or circulation.
2. Sketch	Draw the vector, curve, planar region, solid, or surface whenever possible.
3. Setup	Write the parameterization, bounds, differential element, and orientation.
4. Calculate	Differentiate/integrate carefully and simplify.
5. Check	Verify units, signs, limits, and whether the result answers the question.

Contents

- 1. Vectors and three-dimensional geometry
- 2. Vector-valued functions and space curves
- 3. Double and triple integrals
- 4. Conic sections and quadratic surfaces
- 5. Gradient, tangent planes, and optimization
- 6. Vector fields and line integrals
- 7. Surface integrals and integral theorems
- 8. Formula sheet, problem-selection guide, and practice

1. Vectors and three-dimensional geometry

A vector records both magnitude and direction. In component form, $a = (a_1, a_2, a_3)$. A point $P(x, y, z)$ identifies a location; a vector describes a displacement or direction.

Core formulas

$|a| = \sqrt{(a_1^2 + a_2^2 + a_3^2)}$

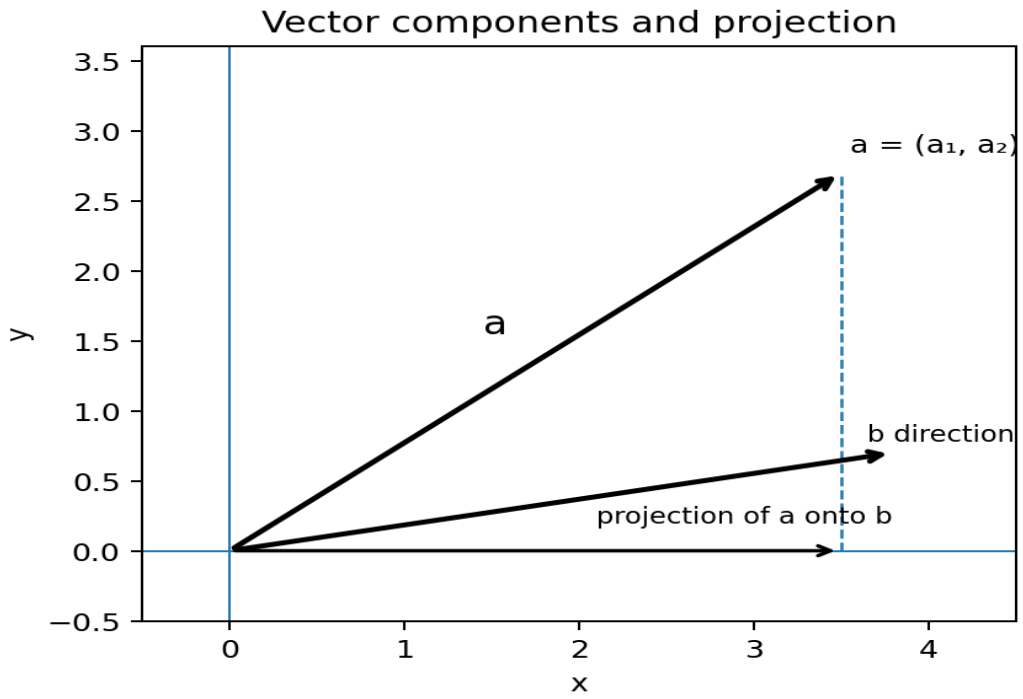
$a \cdot b = a_1b_1 + a_2b_2 + a_3b_3 = |a||b| \cos \theta$

$\cos \theta = (a \cdot b)/(|a||b|)$; perpendicular vectors satisfy $a \cdot b = 0$.

$a \times b = (a_2b_3 - a_3b_2, a_3b_1 - a_1b_3, a_1b_2 - a_2b_1)$

Parallelogram area = $|a \times b|$; parallelepiped volume = $|a \cdot (b \times c)|$.

$\text{proj}_b\, a = [(a \cdot b)/|b|^2]b$; scalar component along b = $(a \cdot b)/|b|$.



Read the picture as a geometric reminder: a vector can be decomposed into components, while a projection keeps only the part parallel to a chosen direction. The projection formula requires $|b|^2$ in the denominator unless b is a unit vector.

Lines and planes

Line: $r(t) = r_0 + tv$.

Plane: $n \cdot (r - r_0) = 0$, equivalently $ax + by + cz = d$.

To find a plane through three non-collinear points, form two direction vectors in the plane and take their cross product to obtain a normal vector.

Worked example

Let $a = (1, 2, 2)$, $b = (2, 0, 1)$. Then $a \cdot b = 4$, $|a| = 3$, $|b| = \sqrt{5}$, so $\cos \theta = 4/(3\sqrt{5})$. For the plane with normal $n = (2, -1, 3)$ through $P(1, 0, 2)$: $2(x-1) - y + 3(z-2) = 0$, hence $2x - y + 3z = 8$.

Common traps: confusing dot product (scalar) with cross product (vector); forgetting the absolute value for area/volume; using a direction vector where a plane needs a normal vector.

2. Vector-valued functions and space curves

A vector function $r(t) = (x(t), y(t), z(t))$ traces a curve through space. Its derivative is tangent to the curve and represents velocity when t is time.

Velocity: $v(t) = r'(t)$; speed: $|r'(t)|$; acceleration: $a(t) = r''(t)$.

Arc length: $L = \int_a^b |r'(t)|\, dt$.

Curvature: $\kappa = |r'(t) \times r''(t)| / |r'(t)|^3$, when $|r'(t)| \neq 0$.

Worked setup

For $r(t) = (t, t^2, 0)$, $0 \leq t \leq 1$, $r'(t) = (1, 2t, 0)$, so speed $= \sqrt{(1 + 4t^2)}$. Therefore $L = \int_0^1 \sqrt{(1 + 4t^2)}\, dt$. The setup is complete even if an exact evaluation is not requested.

Reliable procedure

- Differentiate each component separately.
- Take the magnitude to obtain speed; a vector is not a scalar distance.
- Use the exact parameter interval from the question.
- For curvature, calculate both $r' \times r''$ and $|r'|^3$.

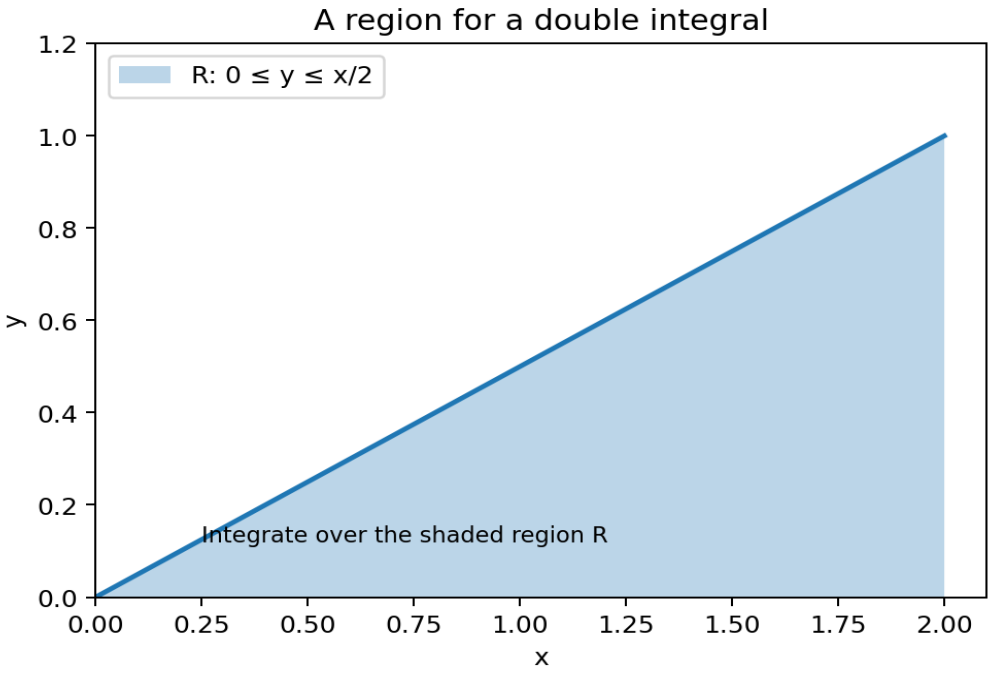
Interpretation: high curvature means the path bends sharply; zero curvature locally indicates no turning at that point, provided the curve is regular.

3. Double and triple integrals

A double integral accumulates a quantity over a planar region R. Integrating 1 gives area. A triple integral accumulates over a solid E; integrating 1 gives volume.

Area(R) = $\iint_R 1\ dA$; Volume(E) = $\iiint_E 1\ dV$.

Double-integral region and bounds

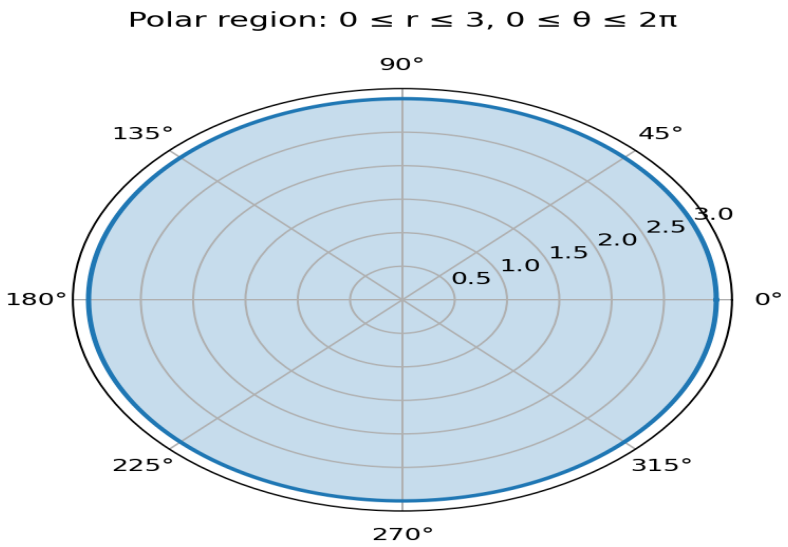


For the shaded triangular region, one possible description is $0 \le x \le 2$ and $0 \le y \le x/2$. Thus $\iint_R f(x,y)dA = \int_0^2 \int_0^{x/2} f(x,y)\ dy\ dx$. Reversing the order requires describing the same region using the other variable first.

Coordinate systems and differential elements

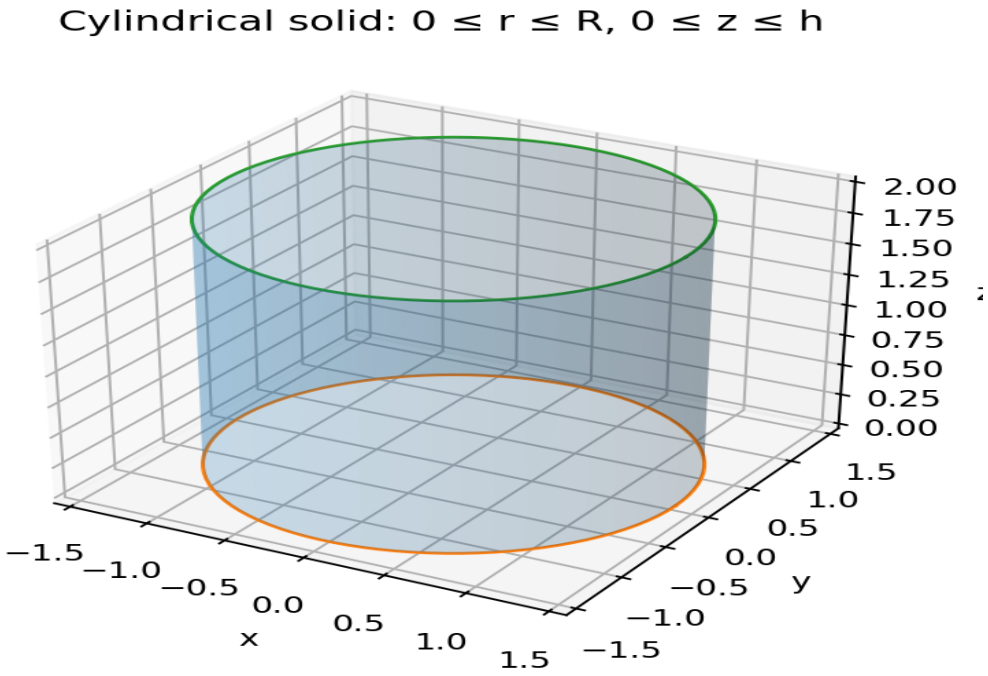
System	Coordinate conversion	Differential element
Rectangular	x, y, z	dA = dx dy; dV = dx dy dz
Polar	$x = r\cos\theta$; $y = r\sin\theta$	$dA = r\ dr\ d\theta$
Cylindrical	$x = r\cos\theta$; $y = r\sin\theta$; z = z	$dV = r\ dz\ dr\ d\theta$ (order may vary)
Spherical	$x = \rho\sin\varphi\cos\theta$; $y = \rho\sin\varphi\sin\theta$; $z = \rho\cos\varphi$	$dV = \rho^2\sin\varphi\ d\rho\ d\varphi\ d\theta$

In this common spherical convention, ρ is distance from the origin, θ is measured in the xy-plane, and φ is measured down from the positive z-axis. Check the convention used by your course.



For a full disk of radius 3, polar bounds are $0 \le r \le 3$ and $0 \le \theta \le 2\pi$. The factor r is essential: $dA = r\,dr\,d\theta$.

Triple integral: visualizing a solid



For a cylinder of radius R and height h, use $0 \le r \le R$, $0 \le \theta \le 2\pi$, and $0 \le z \le h$. Its volume setup is $\int_0^{2\pi} \int_0^R \int_0^h r\,dz\,dr\,d\theta = \pi R^2 h$, with the order adjusted to match the differentials.

Jacobian and applications

$dA = |\partial(x,y)/\partial(u,v)|\,du\,dv$; $dV = |\partial(x,y,z)/\partial(u,v,w)|\,du\,dv\,dw$.

The absolute Jacobian determinant measures local area/volume scaling under a coordinate change. Applications include mass, average value, area, and volume.

Exam checklist: sketch the region/solid; choose coordinates that match its symmetry; set bounds; include the correct Jacobian; check whether the bounds cover the object exactly once.

4. Conic sections and quadratic surfaces

Conic sections in the plane

Curve	Standard form / clue
Circle	$(x-h)^2 + (y-k)^2 = r^2$; fixed radius.
Ellipse	$(x-h)^2/a^2 + (y-k)^2/b^2 = 1$; closed oval.
Parabola	$(x-h)^2 = 4p(y-k)$ or $(y-k)^2 = 4p(x-h)$; one variable is not squared.
Hyperbola	$(x-h)^2/a^2 - (y-k)^2/b^2 = 1$ (or reversed); opposite signs, two branches.

Translation and rotation

Translation shifts the origin. Example: $x^2 - 6x + y^2 + 4y = 12$ becomes $(x-3)^2 + (y+2)^2 = 25$ by completing squares, so the centre is $(3, -2)$ and radius is 5.

Rotation changes axis orientation and can remove an xy term. For $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$, a common relation is $\tan(2\theta) = B/(A-C)$, subject to the chosen convention and special cases.

Quadric surfaces

Surface	Typical form	Recognition
Ellipsoid	$x^2/a^2 + y^2/b^2 + z^2/c^2 = 1$	Closed and bounded.
One-sheet hyperboloid	$x^2/a^2 + y^2/b^2 - z^2/c^2 = 1$	Two positive squared terms, one negative.
Two-sheet hyperboloid	$-x^2/a^2 - y^2/b^2 + z^2/c^2 = 1$	One positive squared term, two negative.
Elliptic paraboloid	$z = x^2/a^2 + y^2/b^2$	Bowl-shaped.
Cone	$z^2 = x^2/a^2 + y^2/b^2$	Vertex at origin in this form.
Cylinder	$x^2/a^2 + y^2/b^2 = 1$	One variable is absent.

To identify a surface, set one variable equal to a constant and inspect the cross-section. Compare it with a standard circle, ellipse, parabola, or hyperbola.

Memory aid: the signs in the squared terms often reveal the family, but the constants and the variable isolated on one side also matter.

5. Gradient, tangent planes, and optimization

For a scalar field f(x,y,z), the gradient collects the first partial derivatives and points in the direction of steepest local increase.

$\nabla f = (f_x, f_\gamma, f_z)$.

Directional derivative: $D_u f = \nabla f \cdot u$, where $|u| = 1$.

If a direction v is given, normalize it first: $u = v/|v|$.

Worked example

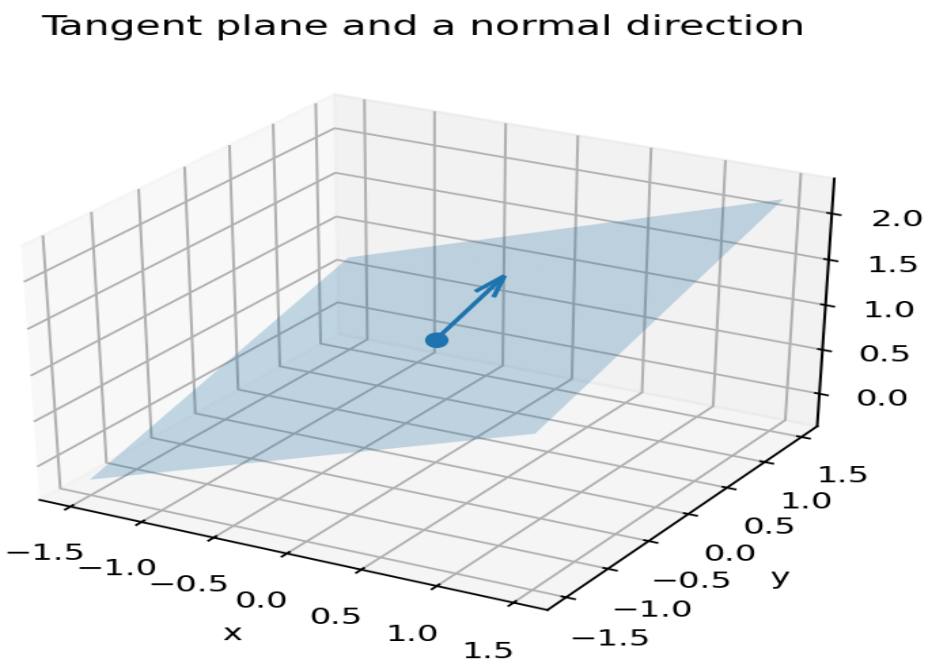
For $f(x,y,z) = x^2 + yz$, $\nabla f = (2x, z, y)$. At P(1,2,3), $\nabla f(P) = (2,3,2)$. In the direction v = (2,0,0), the unit vector is u = (1,0,0), so $D_u f = 2$.

Tangent plane and normal line

For a level surface F(x,y,z) = c, a normal vector at P is $\nabla F(P)$.

Tangent plane: $\nabla F(P) \cdot (r - P) = 0$.

Normal line: $r(t) = P + t\nabla F(P)$.



The gradient is perpendicular to the tangent plane of a regular level surface. The sketch shows a plane, a point on it, and a normal direction.

Lagrange multipliers

$\nabla f = \lambda \nabla g$, together with g(x,y,z) = c.

- Write the constraint.
- Compute both gradients.
- Solve the component equations with the constraint.
- Evaluate f at every candidate.
- Compare all candidate values and check boundaries/regularity when relevant.

The equations give candidates, not automatically a final maximum or minimum. Compare objective values and ensure the constraint set has been handled completely.

6. Vector fields and line integrals

A scalar field assigns a number to each point; a vector field assigns a vector, for example F = (P,Q,R). A line integral accumulates a field along an oriented curve.

For r(t), $a \le t \le b$: $\int_C F \cdot dr = \int_a^b F(r(t)) \cdot r'(t)\, dt$.

Worked example

Let F = (y,x) and r(t) = (t,t^2), $0 \le t \le 1$. Then F(r(t)) = (t^2,t), r′(t) = (1,2t), and the dot product is $3t^2$. Therefore $\int_C F \cdot dr = \int_0^1 3t^2\, dt = 1$.

Conservative fields

If F = ∇f, then $\int$ from A to B $F \cdot dr = f(B) - f(A)$.

The integral depends only on endpoints, not the path. In 2D, for F = (P,Q), the test $P_y = Q_x$ is useful on a suitable simply connected domain with continuous partial derivatives. It is not a universal guarantee if the domain has holes or regularity conditions fail.

Green's theorem

$\oint_C P\, dx + Q\, dy = \iint_R (Q_x - P_y)\, dA$.

Use it for a closed planar boundary with appropriate smoothness. Counterclockwise is the positive orientation. Check that the curve is closed and bounds the region in question.

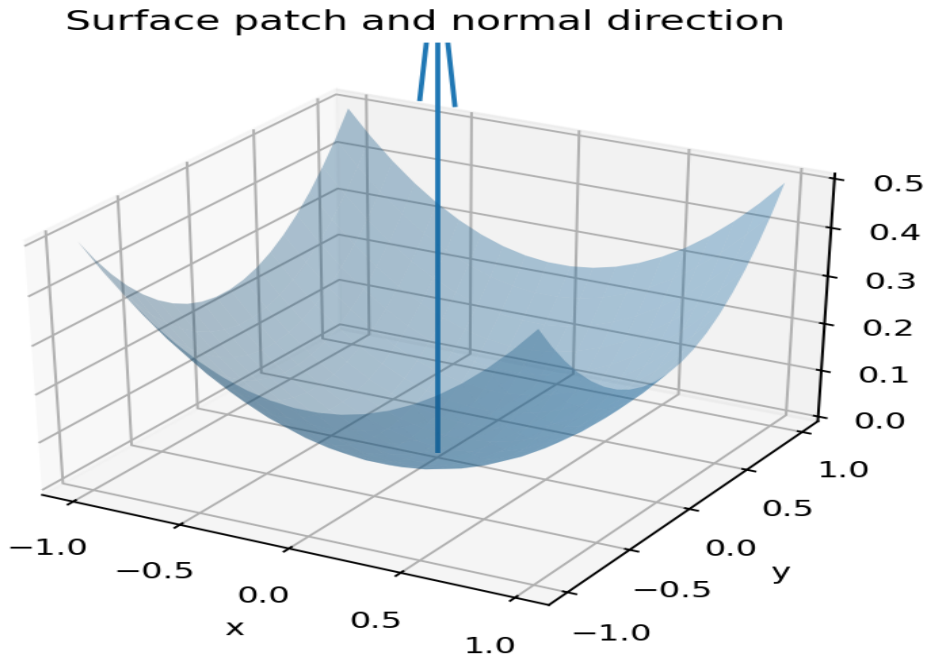
7. Surface integrals and integral theorems

For a parameterized surface r(u,v), tangent vectors r_u and r_v span the tangent plane. Their cross product is normal to the surface.

Surface element: $dS = |r_u \times r_v|\, du\, dv$.

Scalar surface integral: $\iint_S f\, dS = \iint f(r(u,v))\, |r_u \times r_v|\, du\, dv$.

Flux: $\iint_S F \cdot n\, dS$; its sign depends on orientation.



A normal vector indicates the direction perpendicular to a surface patch. For flux, the chosen normal direction matters; reversing it reverses the sign.

Divergence and curl

For F = (P,Q,R): $\operatorname{div} F = \nabla \cdot F = P_x + Q_\gamma + R_z$.

$\operatorname{curl} F = \nabla \times F = (R_\gamma - Q_z,\ P_z - R_x,\ Q_x - P_\gamma)$.

Divergence measures local net outward flow; curl measures local rotational tendency. Keep the component order consistent.

Three major integral theorems

Theorem	Formula	Use it when…
Green	$\oint_c P\, dx + Q\, dy = \iint_r (Q_x - P_\gamma) dA$	A closed planar curve bounds a suitable region.
Divergence	$\oiint_S F \cdot n\, dS = \iiint_e (\nabla \cdot F) dV$	S is a closed surface enclosing a solid; use outward normal.
Stokes	$\oint_c F \cdot dr = \iint_s (\nabla \times F) \cdot n\, dS$	A space curve bounds an oriented surface; orientations agree.

Right-hand rule: for Stokes' theorem, the positive boundary direction and chosen surface normal must match. Green's theorem uses the positive (counterclockwise) boundary orientation. The divergence theorem uses outward normal on a closed surface.

8. Formula review and exam strategy

Topic	Formula to recognize
Magnitude	$\|a\| = \sqrt{(a_1^2 + a_2^2 + a_3^2)}$
Dot product	$a \cdot b = \|a\|\|b\| \cos \theta$
Cross product area	Area = $\|a \times b\|$
Scalar triple product	Volume = $\|a \cdot (b \times c)\|$
Line / plane	$r = r_0 + tv$; $n \cdot (r - r_0) = 0$
Arc length	$L = \int \|r'(t)\| \, dt$
Curvature	$\kappa = \|r' \times r''\| / \|r'\|^3$
Polar area element	$dA = r \, dr \, d\theta$
Cylindrical volume element	$dV = r \, dz \, dr \, d\theta$
Spherical volume element	$dV = \rho^2 \sin \varphi \, d\rho \, d\varphi \, d\theta$
Directional derivative	$D_u f = \nabla f \cdot u$, $\|u\| = 1$
Tangent plane	$\nabla F(P) \cdot (r - P) = 0$
Lagrange multipliers	$\nabla f = \lambda \nabla g$; $g = c$
Green	$\oint P \, dx + Q \, dy = \iint (Q_x - P_\gamma) dA$
Divergence theorem	$\oiint F \cdot n \, dS = \iiint \text{div}(F) dV$
Stokes	$\oint F \cdot dr = \iint (\text{curl } F) \cdot n \, dS$

Which method should I choose?

If the problem asks for…	First move
Angle / perpendicularity	Dot product; normalize only if a unit direction is required.
Area / volume from vectors	Cross product magnitude / absolute scalar triple product.
Curve length	Differentiate r(t), compute $\|r'(t)\|$, integrate over the parameter interval.
Area / volume integral	Sketch bounds; use rectangular, polar, cylindrical, or spherical coordinates.
Tangent plane to F=c	Compute ∇F at the point and use it as the normal.
Line integral	Parameterize the curve, substitute F(r(t)), dot with r′(t).
Closed planar line integral	Check whether Green's theorem simplifies it.
Flux through closed surface	Check the divergence theorem.
Circulation around a space curve	Check Stokes' theorem and orientation.

Practice questions

- 1. Find the angle between a = (1, −1, 2) and b = (2, 1, 0).
- 2. Find the plane through P(1, 0, 2) with normal (2, −1, 3).
- 3. For r(t) = (t, t^2, t^3), find r′(t), speed, and the arc-length setup on $0 \le t \le 1$.
- 4. Evaluate $\iint_R (2x - y)\, dA$ over $0 \le x \le 1$, $0 \le y \le 2$.
- 5. Write polar bounds for $x^2 + y^2 \le 9$ and set up the area integral.
- 6. Classify $x^2/4 + y^2/9 - z^2 = 1$.
- 7. Find ∇f for $f = x^2y + z^3$ and evaluate at (1, 2, −1).
- 8. For F = (2x, 2y), test conservativeness and find a potential function.
- 9. State the orientation condition for Green's and Stokes' theorems.
- 10. Explain when the divergence theorem is more efficient than direct flux integration.

Answer check / key

- 1. a·b = 1; |a| = √6, |b| = √5, so $\cos\theta = 1/\sqrt{30}$.
- 2. 2x − y + 3z = 8.
- 3. r′(t) = (1, 2t, $3t^2$); speed = $\sqrt{1 + 4t^2 + 9t^4}$; $L = \int_0^1 \sqrt{1 + 4t^2 + 9t^4}\, dt$.
- 4. $\int_0^1 \int_0^2 (2x - y)\, dy\, dx = 0$.
- 5. $0 \le r \le 3$, $0 \le \theta \le 2\pi$; area = $\int_0^{2\pi} \int_0^3 r\, dr\, d\theta = 9\pi$.
- 6. One-sheet hyperboloid (two positive squared terms, one negative).
- 7. ∇f = (2xy, x^2, $3z^2$); at (1,2,−1): (4,1,3).
- 8. $P_y = 0 = Q_x$; potential $f = x^2 + y^2 + C$.
- 9. Green: positive boundary is counterclockwise. Stokes: boundary orientation agrees with the chosen normal by the right-hand rule.
- 10. When the surface is closed and divergence is simpler to integrate over the enclosed volume than flux over the surface.

Final exam checklist

- Geometry: correct point, direction, normal, curve, region, and orientation?
- Coordinates: bounds cover the region exactly once, and the Jacobian is included?
- Vectors: unit direction used when needed?
- Theorems: closure, smoothness, and orientation requirements satisfied?
- Result: correct sign, simplified, and stated in the requested form?

Reference note: This is an original study summary based on the topic sequence of the supplied nine-page PDF and standard multivariable-calculus methods associated with Howard Anton, Irl Bivens & Stephen Davis, Calculus: Early Transcendentals, and James Stewart's calculus texts. For full derivations, use the edition assigned in your course and your instructor's conventions.